



# Quantum-inspired dynamical models on quantum and classical annealers

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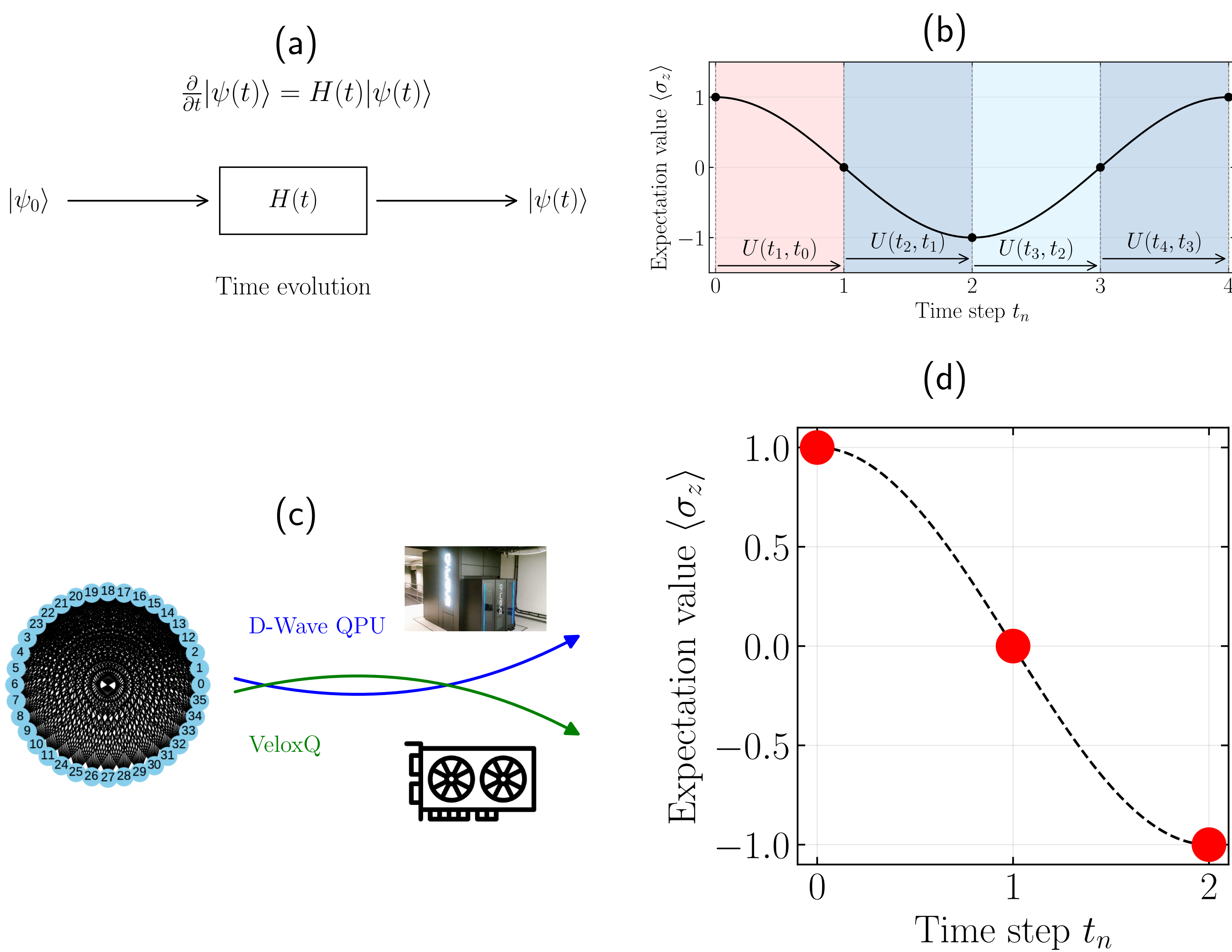
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## Introduction

We propose a practical, physics-inspired benchmarking suite to challenge both quantum and classical computers by mapping real-time quantum dynamics to a common optimization format.



## Methods

### Quantum Annealing

#### ► Driver Hamiltonian:

$$|+\rangle^{\otimes N} = \frac{1}{\sqrt{2^N}} \sum_{x \in \{0,1\}^N} |x\rangle$$

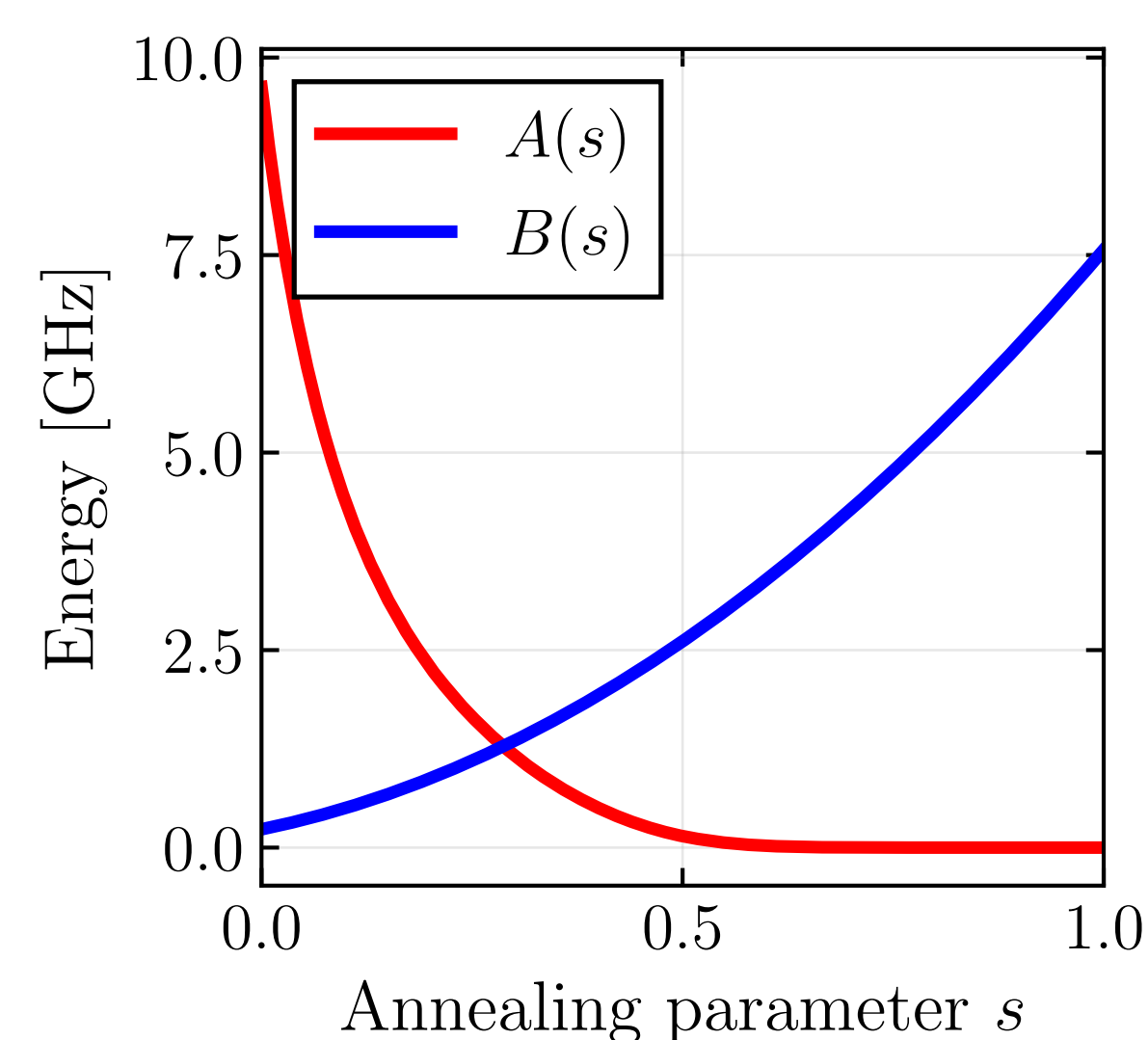
where  $N$  is the number of qubits.

#### ► Problem Hamiltonian:

$$H_P = \sum_{i=1}^N h_i \hat{\sigma}_z^{(i)} + \sum_{i < j} J_{ij} \hat{\sigma}_z^{(i)} \hat{\sigma}_z^{(j)}.$$

#### ► System evolution:

$$H_{\text{Ising}} = \frac{A(s)}{2} H_D + \frac{B(s)}{2} H_P.$$



### VeloxQ

VeloxQ is a fast and efficient, physics-inspired solver based on the theory of dynamical systems, designed for solving QUBO problems.

## Problem Description

### ► Quantum State Evolution

$$|\Psi\rangle = \sum_{n=0}^{N-1} |t_n\rangle \otimes |\psi(t_n)\rangle.$$

### ► Annihilated by Hermitian clock operator

$$\mathcal{C} = \sum_{n=0}^{N-2} \left( |t_{n+1}\rangle \langle t_{n+1}| \otimes I - |t_{n+1}\rangle \langle t_n| \otimes U_n + \text{h.c.} \right)$$

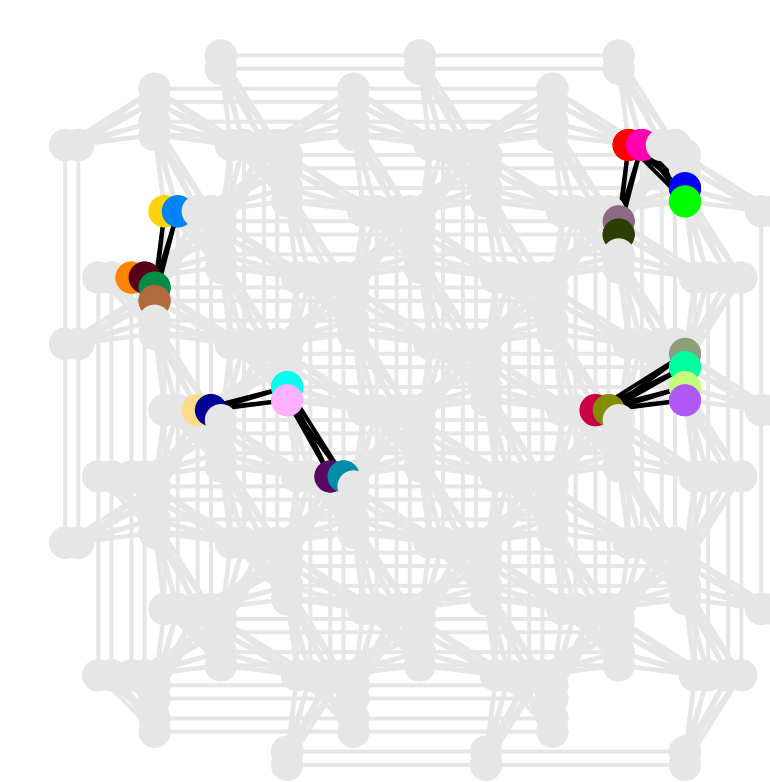
|                     |                             |                             |                             |
|---------------------|-----------------------------|-----------------------------|-----------------------------|
| $2\hat{I}$          | $\hat{U}^\dagger(t_1, t_0)$ | 0                           | 0                           |
| $\hat{U}(t_1, t_0)$ | $2\hat{I}$                  | $\hat{U}^\dagger(t_2, t_1)$ | 0                           |
| 0                   | $\hat{U}(t_2, t_1)$         | $2\hat{I}$                  | $\hat{U}^\dagger(t_3, t_2)$ |
| 0                   | 0                           | $\hat{U}(t_3, t_2)$         | $\hat{I}$                   |

Clock Hamiltonian  $\mathcal{C}$

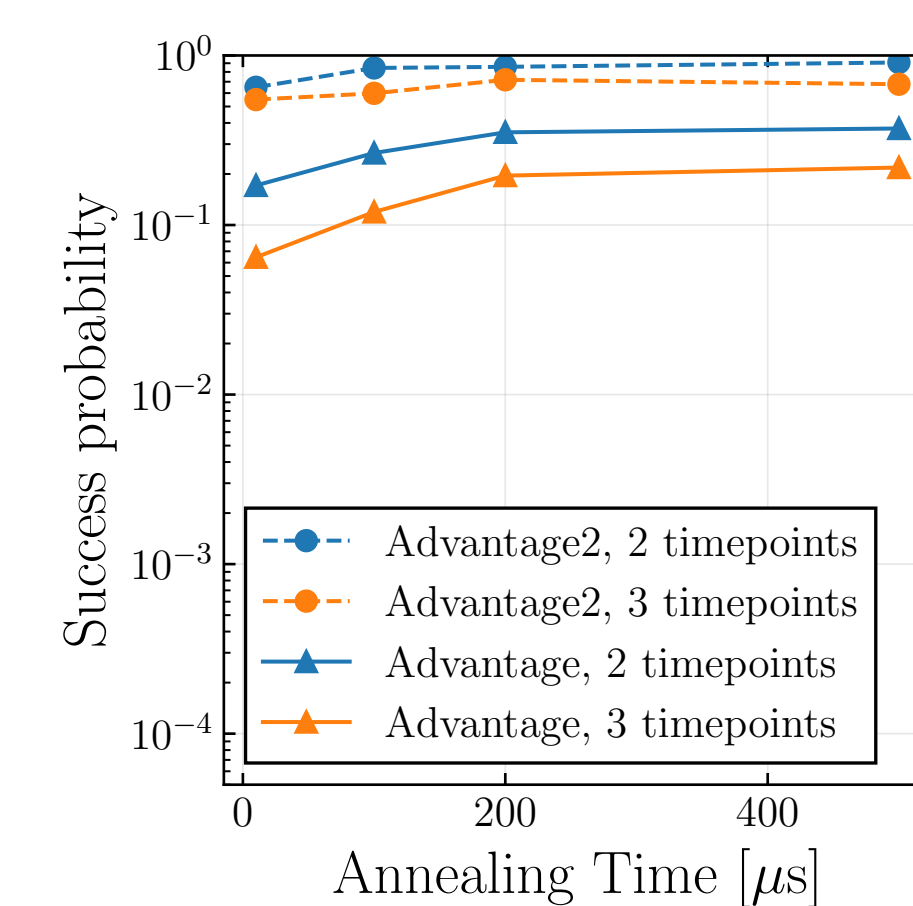
### ► Finally minimize

$$f(\mathbf{x}) = \frac{1}{2} \langle \mathbf{x} | \mathcal{A} | \mathbf{x} \rangle - \langle \mathbf{x} | \phi \rangle, \text{ with } \mathcal{A} := \mathcal{C} + |t_0\rangle \langle t_0| \otimes I$$

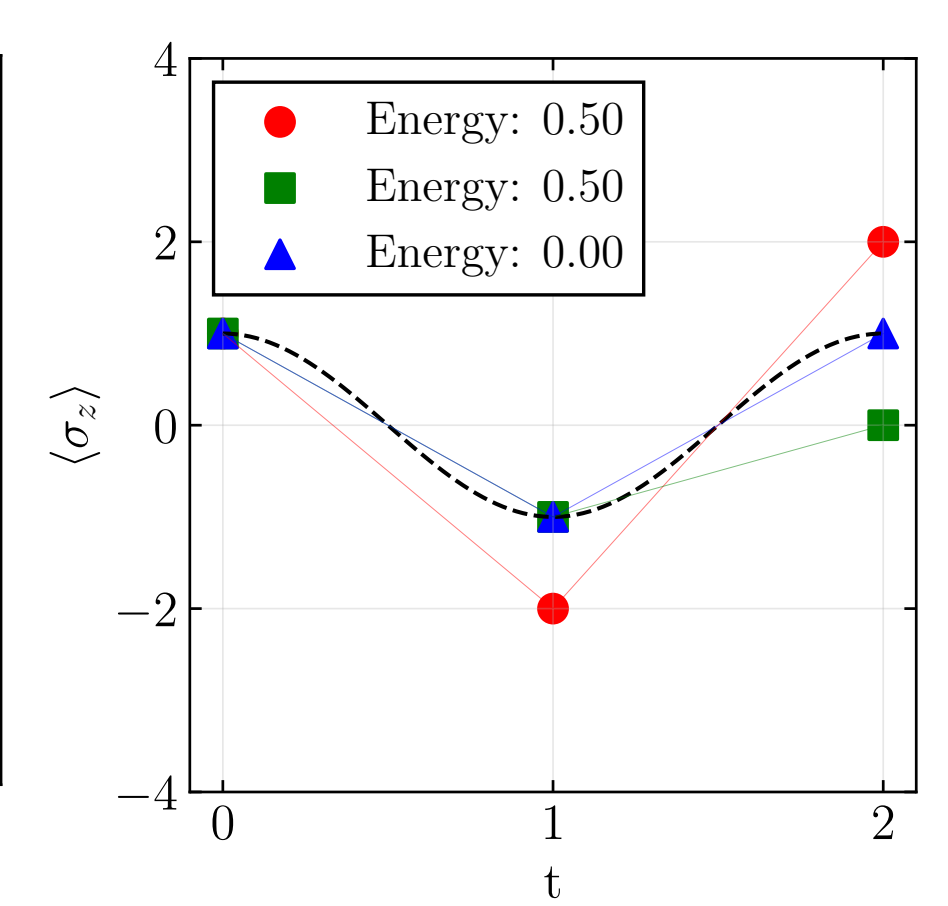
### Solving process



(a) Example embedding on QPU



(b) Success rates for systems  $\hat{H}_2$



(c) Expectation value  $\langle \sigma_z \rangle$

## Results

1. Data Collection.
2. Signal Processing.
3. Result Verification.

## Conclusion

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